SURVEYING & LEVELLING · NOTE SL13

Contours and Topographic Surveys — Grid and Spot-Level Methods

Course topics: CAE1.3 L10, SWCE1.4 L13

Purpose of a topographic survey

A topographic survey collects the data for a topographic map, which shows the relief of the land, its dimensions, natural features (streams, gullies, ditches) and man-made features (bridges, culverts, buildings, roads, fences). The most practical way to show topography is a **contour map**.

Contours

A contour is an imaginary line of constant elevation on the ground, shown on maps as a contour line. Successive contour lines differ by a fixed vertical distance, the **contour interval**, which usually varies from 20 to 200 cm depending on the terrain.

Characteristics of contour lines

- Every point on a contour has the same elevation.
- Close spacing means steep ground; wide spacing means flat ground. Uniform slopes give uniformly spaced contours; plane surfaces give straight, parallel contours.
- Contours run perpendicular to the direction of steepest slope.
- Contours never end abruptly — they close on themselves inside or outside the map.
- A closed contour marks a **summit** if the higher values are inside and a **depression** if the lower values are inside.
- Contours do not cross, except at an overhanging cliff.
- Across a **ridge** they form V (or U) shapes with the higher values inside; across a **valley** they form sharp Vs with the lower values inside, pointing upstream.

Uses of contour maps

- Judge whether terrain is flat, undulating or mountainous; guide land-improvement planning.
- Delineate watershed boundaries.
- Plan irrigation water distribution systems and land drainage systems.
- Prepare cost estimates for land development and soil and water conservation plans.
- Estimate runoff for waterways, gully control structures and farm pond spillways.
- Choose economical alignments for irrigation channels, drainage channels and roads.

Drainage area of a point is the area that contributes flow through it. The volume of water stored between two contours = $(A_1 + A_2)/2 \times$ contour interval.

Methods of topographic surveying

Method	Notes
Grid system	Most commonly used; more accurate but slower; needs only a level and a chain or tape
Plane table and alidade	Graphical; plots in the field
Transit (theodolite) and stadia	Angles and stadia distances; quick over rough ground

Grid survey

1. **Reconnoitre** the area first and fix a bench mark — of known elevation, or assumed (0.00 m or 100.00 m).
2. Lay out one or more base lines, then lines parallel and perpendicular to them at fixed intervals, forming squares. Grid sizes vary from 3 m to 30 m or more; 15 m, 20 m and 30 m are common.
3. Peg the corners and find their elevations by profile levelling. Keep the distance between rod readings within 100 m (maximum), preferably 70 m.
4. Take extra points wherever the ground is irregular.
5. Compute the RLs of the grid points, enter them in a table or plot them on the base map, and draw the contours by interpolation.

Range of land slope	Recommended contour interval
0 – 1%	5 – 15 cm
1 – 2%	15 – 30 cm
2 – 5%	30 – 60 cm
5 – 10%	60 – 150 cm

Base-line and spot-level method

Run base lines from bench marks and take spot levels on both sides, measuring distances by tape, chain or stadia — essentially profile levelling. It suits buried pipelines and on-farm irrigation and drainage works.

Drawing the contour map

- **Interpolation** spaces the contours proportionately between plotted points — by eye or by arithmetic; it takes skill and judgment.
- Draw only elevations that are multiples of the contour interval, as smooth freehand lines of uniform width; draw every fifth contour thicker and leave gaps for the elevation labels.
- Show ground features with standard symbols, a north arrow on each map, soil logs with symbols and groundwater levels with their dates; verify legal descriptions for approved works.
- Pick the scale for the detail needed — e.g. 1 cm = 10 m; use a bar scale where drawings may be reduced.
- Put a **title block** in the lower right-hand corner: subject of the drawing, project name and implementing agency, agency preparing the drawings, revenue sub-division, district and state, designers, drafters and checkers, signatures with dates and drawing number.

Topographic maps serve from planning to final construction: lay out irrigation and drainage systems, conservation structures and farm roads on the map, pick the best layout and improve the alignment before building. A layout may be drafted at a larger scale and reduced cartographically. On site, transfer it to the ground with horizontal and vertical controls, laying base lines and details with minimum instrument time.

Watershed location map

It shows the contributing watershed boundary, the location of the proposed works and the data needed for analysis and design:

1. average slope
2. streams, drains, tanks and depressions
3. land-use classification
4. area of each soil type
5. location of works (check dams, channels)
6. bridges, culverts and other structures
7. property ownership
8. roads, buildings and pipelines
9. legal section and centre details

Prepare it by marking details on a topographic map, or from aerial photographs — traced, enlarged, or enlarged with a pantograph.

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. (Lessons 11–13) Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Closely spaced contours indicate

(a) flat ground (b) steep ground
(c) a depression (d) a valley floor only

Q2. Contour lines cross each other only at

(a) a ridge (b) a valley
(c) an overhanging cliff (d) a saddle

Q3. A closed contour with higher values inside indicates

(a) a depression (b) a summit
(c) a valley (d) a cliff

Q4. Recommended contour interval for land with 1–2% slope is

(a) 5–15 cm (b) 15–30 cm
(c) 30–60 cm (d) 60–150 cm

Q5. The most common method of topographic survey for farm work is the

(a) grid system (b) plane table
(c) stadia traverse (d) aerial survey

Q6. Every ___ contour is drawn thicker

(a) second (b) third
(c) fifth (d) tenth

Answers

Q1 — (b) steep ground

Q2 — (c) an overhanging cliff

Q3 — (b) a summit

Q4 — (b) 15–30 cm

Q5 — (a) grid system

Q6 — (c) fifth